

## THE MOLE CONCEPT

- Moles are SI units expressing the quantity of a substance. They are often abbreviated as “mol”.
- When balancing chemical equations, the **stoichiometric coefficients (the numbers attached to each chemical formula)** represent the **lowest ratio of the number of moles needed to complete the reaction**.

## BALANCING CHEMICAL EQUATIONS

- Based on the *Law of Conservation of Mass* by *Antoine Lavoisier*, which states that matter cannot be created nor destroyed.
- Any chemical reaction in the form  $A + B \rightarrow C + D$  or  $A + B \rightarrow C$ , when balanced, has coefficients attached to the chemical formulas. These coefficients are what we call **stoichiometric coefficients**. These convert these reactions into the forms  $aA + bB \rightarrow cC + dD$  or  $aA + bB \rightarrow cC$ .

**Example 1:** Balance the chemical equation  $\text{Xe} + \text{F}_2 \rightarrow \text{XeF}_6$  (Xenon reacts with Fluorine gas to produce Xenon Hexafluoride).

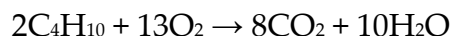
### Solution:

Notice that, on the reactant side, there is exactly one Xe atom and 2 F atoms. On the right, there is also one Xe atom, but there are 6 F atoms. What we must do is to find a way for those 2 F atoms on the reactant side to match the 6 F atoms on the product side. In this case, we must multiply  $\text{F}_2$  by 3 to create 6 F atoms. All-in-all, the final chemical equation is  $\text{Xe} + 3\text{F}_2 \rightarrow \text{XeF}_6$ .

**Example 2:** Balance the chemical equation for the combustion of butane, given by  $\text{C}_4\text{H}_{10} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ .

### Solution:

Similarly, we count the number of each atom on the reactant and the product side. Take note that in balancing, H and O atoms are the **least priority**. We first balance the C atoms by multiplying  $\text{CO}_2$  by 4. Then, we balance H atoms by multiplying 5 to  $\text{H}_2\text{O}$ . Lastly, there will be a total of 13 O atoms on the product side. So, we must multiply the  $\text{O}_2$  on the reactant side by  $\frac{13}{2}$ . Since this is not a whole number, we multiply everything by 2 to cancel the denominator. All-in-all, we get:



## MOLE RATIOS

As stoichiometric coefficients represent the **lowest ratio of the number of molecules needed for a complete reaction**, if we know how many moles of one substance are going to be used in the reaction, we can also know how many moles of all the other substances are needed. Take the following example.

**Example 3:** Given the chemical equation  $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ , how many moles of oxygen gas are needed to completely synthesize 6 moles of hydrogen gas into water? How many moles of water are then formed?

### Solution:

As the given equation is already balanced, we simply take mole ratios. The mole ratios are written as follows:

$$\frac{2 \text{ mol H}_2}{1 \text{ mol O}_2}$$

$$\frac{2 \text{ mol H}_2}{2 \text{ mol H}_2\text{O}} = \frac{1 \text{ mol H}_2}{1 \text{ mol H}_2\text{O}}$$

$$\frac{1 \text{ mol O}_2}{2 \text{ mol H}_2\text{O}}$$

Then, to find the number of moles of each substance, we must strictly adhere to these mole ratios. For instance, to find the number of moles of oxygen, we simply state:

$$\frac{6 \text{ mol H}_2}{x \text{ mol O}_2} = \frac{2 \text{ mol H}_2}{1 \text{ mol O}_2}$$

We can cancel the units to simplify this into a simple ratio problem.

$$\frac{6}{x} = \frac{2}{1}$$

$$6 = 2x$$

$$3 = x$$

So, **3 moles of oxygen gas (O<sub>2</sub>)** are needed. Likewise, we can do the same for finding the total product. Since H<sub>2</sub>:H<sub>2</sub>O is a 1:1 ratio, we can simply state that we also produce **6 moles of water (H<sub>2</sub>O)**.

**Example 4:** John combusts 4 moles of butane, which follows the balanced chemical equation  $2\text{C}_4\text{H}_{10} + 13\text{O}_2 \rightarrow 8\text{CO}_2 + 10\text{H}_2\text{O}$ . How many moles of oxygen gas ( $\text{O}_2$ ) are needed to complete this combustion? How many moles of carbon dioxide ( $\text{CO}_2$ ) and water ( $\text{H}_2\text{O}$ ) are formed?

**Solution:**

Again, we can simply take the ratios. To not complicate further, let us use simple ratio notation. We know that the ratio of  $\text{C}_4\text{H}_{10}$  to  $\text{O}_2$  is 2:13, so as 4 moles of butane are used, we state  $4:x = 2:13$ . Solving, we find that  $x = 26$ , so we need **26 moles of oxygen gas** to complete this reaction.

Similarly, for the number of  $\text{CO}_2$  moles, we know that the ratio of  $\text{C}_4\text{H}_{10}$  to  $\text{CO}_2$  moles is 2:8 or 1:4. Setting  $4:y = 1:4$ , we find that  $y = 16$ . So, we produce **16 moles of carbon dioxide**.

Lastly, the ratio for  $\text{C}_4\text{H}_{10}$  and  $\text{H}_2\text{O}$  is 2:10, or 1:5. So, we set  $4:z = 1:5$ . Solving for  $z$ , we find that  $z = 20$ , meaning that we form **20 moles of water**.